RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Sunday, 30 August 2026, 10:51 AM
State	Finished
Completed on	Sunday, 30 August 2026, 11:49 AM
Time taken	58 mins 39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int a[], int low, int high, int x)
{
    if (low > high)
        return -1;

    if (x < a[low])
        return -1;

    if (x >= a[high])
        return a[high];

    int mid = (low + high) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] < x)
    {
        int result = findFloor(a, mid + 1, high, x);
        if (result == -1)
            return a[mid];
        return result;
    }
    else
    {
        return findFloor(a, low, mid - 1, x);
    }
}

int main()
{
    int n, x;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d", findFloor(a, 0, n - 1, x));

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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